EXPERIMENT NO: 14 **Date: _____**

IMPLEMENTATION OF INNER JOIN

AIM

To implement **INNER JOIN** in SQL to retrieve related data from multiple tables using common attributes.

OBJECTIVES

1. Create the **STUDENT**, **BOOK**, **RESERVED**, and **ACCESSED** tables with appropriate primary and foreign keys.
2. Insert sample records into all the tables.
3. Retrieve the student name, marks, reserved book ID, and reservation date using an **INNER JOIN** between the STUDENT and RESERVED tables.
4. Retrieve the student ID, student name, book ID, and book name by joining the STUDENT, ACCESSED, and BOOK tables.
5. Display only the records where the book access date is between **01-09-2023** and **20-09-2023**.

PROGRAM

1. Create STUDENT Table

```
CREATE TABLE STUDENT (
    STD_ID INT PRIMARY KEY,
    NAME VARCHAR(50),
    MARK INT
);
```

2. Create BOOK Table

```
CREATE TABLE BOOK (
    BOOK_ID INT PRIMARY KEY,
    BNAME VARCHAR(50)
);
```

3. Create RESERVED Table

```
CREATE TABLE RESERVED (
    RES_ID INT PRIMARY KEY,
    STD_ID INT,
    BOOK_ID INT,
    RDATE DATE,
    FOREIGN KEY (STD_ID) REFERENCES STUDENT(STD_ID),
    FOREIGN KEY (BOOK_ID) REFERENCES BOOK(BOOK_ID)
);
```

4. Create ACCESSED Table

```
CREATE TABLE ACCESSED (
    ACCESS_ID INT PRIMARY KEY,
    STD_ID INT,
    BOOK_ID INT,
    ADATE DATE,
    FOREIGN KEY (STD_ID) REFERENCES STUDENT(STD_ID),
    FOREIGN KEY (BOOK_ID) REFERENCES BOOK(BOOK_ID)
);
```

5. Insert Records into STUDENT

```
INSERT INTO STUDENT VALUES (101,'Arjun',85);
INSERT INTO STUDENT VALUES (102,'Meera',92);
INSERT INTO STUDENT VALUES (103,'Rahul',78);
INSERT INTO STUDENT VALUES (104,'Sara',88);
INSERT INTO STUDENT VALUES (105,'Anu',90);
```

6. Insert Records into BOOK

```
INSERT INTO BOOK VALUES (201,'Database Management System');
INSERT INTO BOOK VALUES (202,'Computer Networks');
INSERT INTO BOOK VALUES (203,'Operating Systems');
INSERT INTO BOOK VALUES (204,'Data Structures');
INSERT INTO BOOK VALUES (205,'Python Programming');
```

7. Insert Records into RESERVED

```
INSERT INTO RESERVED VALUES (1,101,201,TO_DATE('2023-09-05','YYYY-MM-DD'));
INSERT INTO RESERVED VALUES (2,102,202,TO_DATE('2023-09-08','YYYY-MM-DD'));
INSERT INTO RESERVED VALUES (3,104,205,TO_DATE('2023-09-18','YYYY-MM-DD'));
```

8. Insert Records into ACCESSED

```
INSERT INTO ACCESSED VALUES (1,101,201,TO_DATE('2023-09-03','YYYY-MM-DD'));
INSERT INTO ACCESSED VALUES (2,102,202,TO_DATE('2023-09-10','YYYY-MM-DD'));
INSERT INTO ACCESSED VALUES (3,103,203,TO_DATE('2023-09-22','YYYY-MM-DD'));
INSERT INTO ACCESSED VALUES (4,104,205,TO_DATE('2023-09-15','YYYY-MM-DD'));
```

9. Display Student Name, Mark, Reserved Book ID and Reservation Date

```
SELECT S.NAME,
       S.MARK,
       R.BOOK_ID,
       R.RDATE
FROM STUDENT S
INNER JOIN RESERVED R
ON S.STD_ID = R.STD_ID;
```

10. Display Student ID, Student Name, Book ID and Book Name for Books Accessed Between 01-09-2023 and 20-09-2023

```
SELECT S.STD_ID,
       S.NAME,
       B.BOOK_ID,
       B.BNAME
FROM STUDENT S
INNER JOIN ACCESSED A
ON S.STD_ID = A.STD_ID
INNER JOIN BOOK B
ON A.BOOK_ID = B.BOOK_ID
WHERE A.ADATE BETWEEN
TO_DATE('2023-09-01','YYYY-MM-DD')
AND
TO_DATE('2023-09-20','YYYY-MM-DD');
```

OUTPUT

1. Table Creation

```
Table STUDENT created.

Table BOOK created.

Table RESERVED created.

Table ACCESSED created.
```

2. Record Insertion

```
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.

1 row inserted.
1 row inserted.
```

```
1 row inserted.
1 row inserted.
1 row inserted.

1 row inserted.
1 row inserted.
1 row inserted.

1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
```

3. Student Name, Mark, Reserved Book ID and Reservation Date

NAME	MARK	BOOK_ID	RDATE
Arjun	85	201	05-SEP-23
Meera	92	202	08-SEP-23
Sara	88	205	18-SEP-23

4. Student ID, Student Name, Book ID and Book Name

STD_ID	NAME	BOOK_ID	BNAME
101	Arjun	201	Database Management System
102	Meera	202	Computer Networks
104	Sara	205	Python Programming

RESULT

The **INNER JOIN** queries were executed successfully to retrieve matching records from the **STUDENT**, **BOOK**, **RESERVED**, and **ACCESSED** tables based on common key attributes. The required data was obtained correctly using join conditions and date filtering.